

Rahul Verma

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EDUCATION

University of California, Berkeley

Berkeley, CA

B.S. Electrical Engineering and Computer Sciences, Minor in Mechanical Engineering

Aug. 2026 – May 2029

- **Current Coursework:** EE 64 (Circuits), EE 66 (Signals, Dynamics, & Information), CS 61A (Python)
- **Prior Coursework:** Multivariable Calculus, Linear Algebra, Differential Equations — MiraCosta College

Del Norte High School

San Diego, CA

GPA: 4.46 Weighted; ACT: 36

2022 – 2026

EXPERIENCE

UCSD Supercomputer Center | *Research Intern*

Summer 2024, 2025

- Benchmarked & optimized ML models in PyTorch/TensorFlow across NVIDIA A100/V100 GPUs, finding superior A100 performance on medical/image-generation models and diminishing efficiency returns past 32 CPU cores
- Optimized supercomputer usage for real-time agricultural monitoring and detecting of soybean fields in Iowa
- Presented research poster on the newly designed benchmarking workflow; poster featured on ICICLE research organization website and workflow now used across center

FIRST Tech Challenge – Team 22105 | *Founder, Co-Captain, CAD/Design Lead*

2022 – 2026

- Founded and led a 15-student team across five schools; designed, built, and programmed seven competition robots
- Integrated mechanical, electrical, & software systems, leading to qualifying for the FTC World Championship

Competitive Programming Initiative | *Project Manager for National Nonprofit Organization*

2023 – 2025

- Coordinated 8 ten-week computer science courses for 250+ students to prepare for the USA Computing Olympiad
- Created/taught lessons in competitive programming, managed/onboarded instructors, and handled finances

PROJECTS

Omni-Based Swerve Drive | *CAD, Kinematics, Control Theory*

2024 – 2025

- Designed and built a coaxial swerve-omni drivetrain that increases forward speed past the motor's nominal limit by tilting each wheel's yaw axis, achieving an effect similar to a continuously variable transmission (CVT)
- Developed a velocity-based nonlinear control algorithm, balancing speed against traction loss. Achieved a 1.63x speed boost and won 1st Place in Engineering: Robotics at Greater San Diego Science and Engineering Fair

Telescopic and Pivoting Arm | *CAD, Controls, Mechanism Design*

2025 – 2026

- Engineered a telescoping, pivoting aluminum-tube arm (4 nested pocketed tubes, 2" to 0.5") through 48 design iterations, achieving 4x rigidity and 47% less weight than commercial linear slides with under 5mm of backlash
- Derived a gain-scheduling feedforward controller and custom Newton-Raphson S-curve motion profile to account for the arm's variable moment of inertia, cutting full extension to 0.4s and 90° pivots to 0.25s

LEADERSHIP

Del Norte Algorithmic Coding Club | *President*

2023 – 2026

- Led club of 30+ to 15th nationwide in the American Computer Science League; taught competitive programming, boolean algebra, digital electronics, Regex operations, LISP, and Assembly

AWARDS

FTC Michiana International Premier Event – 3rd Place overall across 96 teams

2025

American Computer Science League – Silver Medalist, Senior Division, top 20 of 2000+

2025

National Merit Finalist – Top 1% of U.S. HS Students on PSAT Exam

2026

AIME Qualifier 2x – Top 5% scorer on American Mathematics Competition examination

2022, 2023

TECHNICAL SKILLS

Programming: Java, Python, C/C++, SQL

Engineering: Mechanical Design, Robotics, PCB Design, 3D Printing, CNC Machining, Systems Integration

CAD: SolidWorks, Onshape, Fusion360, Altium Designer

ML / Computing: PyTorch, TensorFlow, NumPy, Matplotlib, Linux, SLURM